



Name: _____
Class: _____
Date: _____

Mr. Rawson • IB DP Physics

Simple Harmonic Motion 5 – Pendulum Lab — Measuring g

Answer Key

As a class, you are about to measure the strength of gravity using nothing but a length of string, a mass, and a stopwatch. Each pair takes one length; the whole class's results together make the graph.

Key concept

For a simple pendulum swinging through a small angle,

$$T = 2\pi\sqrt{\frac{l}{g}}$$

Squaring both sides straightens the relationship, exactly as you did for the spring: $T^2 = \frac{4\pi^2}{g}l$. A graph of T^2 against l is a straight line through the origin, with gradient $4\pi^2/g$ — so the gradient of *today's* graph hands you a value for g .

Your length

Seven pairs, seven lengths, 0.5 m apart. Find your pair in the table and circle your length.

Pair 1	Pair 2	Pair 3	Pair 4	Pair 5	Pair 6	Pair 7
0.5 m	1.0 m	1.5 m	2.0 m	2.5 m	3.0 m	3.5 m

Method

1. Set up your pendulum at your assigned length l , measured from the pivot to the **centre of the bob**. **Check:** the string itself should measure a little less than l — the bob's own radius makes up the rest.
2. Pull the bob to one side by about $10\text{--}15^\circ$ and release from rest, with no push.
3. Time **10 complete oscillations**, starting your count at zero as the bob is released.
4. Repeat for a total of **5 timings** at your one length.

Your data

Length l / m	Time for 10 oscillations / s					Mean t / s
	1	2	3	4	5	

Your period, $T = \text{mean } t \div 10 = \underline{\hspace{2cm}} \text{ s}$ $T^2 = \underline{\hspace{2cm}} \text{ s}^2$

Report your l and your T^2 to the board once your five timings are done.

The whole class's data

Copy every pair's result from the board — seven points make a graph, one point does not.

Length l / m	0.5	1.0	1.5	2.0	2.5	3.0	3.5
T^2 / s^2							

Processing

1) **Plot** a graph of T^2 (vertical) against l (horizontal) using the class's seven points, and draw the best straight line through them.

[3]

Mark scheme

A straight line through the origin (within experimental scatter), gradient $4\pi^2/g \approx 4.03 \text{ s}^2/\text{m}$ for $g = 9.81 \text{ m/s}^2$. Marks for correctly scaled axes, points plotted accurately, and a single best-fit line (not dot-to-dot).

2) **Determine** the gradient of your line, and use it to calculate g .

[4]

My gradient = $\underline{\hspace{2cm}} \text{ s}^2/\text{m}$

$$g = \frac{4\pi^2}{\text{gradient}} = \underline{\hspace{2cm}} \text{ m/s}^2$$

Mark scheme

Correct rearrangement of $T^2 = (4\pi^2/g)l$ to $g = 4\pi^2/\text{gradient (2)}$, correctly evaluated from the student's own gradient (2). A class data set close to ideal should give a gradient near $4.0 \text{ s}^2/\text{m}$ and therefore g close to 9.8 m/s^2 .

- 3) **Compare** your calculated g with the accepted value, 9.81 m/s^2 . Give a percentage difference.

[2]

Key concept

$$\text{percentage difference} = \frac{|\text{your value} - \text{accepted value}|}{\text{accepted value}} \times 100\%$$

The upright bars mean *take the size of the difference and drop the sign*, so a value that comes out high and one that comes out low by the same amount give the same answer.

Percentage difference: _____ %

Mark scheme

$\left(\frac{|g_{\text{calc}} - 9.81|}{9.81}\right) \times 100\%$, correctly evaluated from the student's own calculated g . A well-run class data set should land within a few percent; a much larger gap is worth revisiting the graph before the arithmetic.

- 4) **Explain** why plotting T^2 against l , rather than T against l , lets a straight line hand you a value for g .

[2]

Mark scheme

T against l is a curve, and a curve has no single gradient to measure. Squaring T turns the relationship into a straight line whose gradient is one fixed number, $4\pi^2/g$, which can be rearranged directly for g .

Before Tuesday**Homework**

Finish this worksheet — the graph, the gradient, your value of g , the percentage difference, and the last question — due at the start of Tuesday's lesson.